

Project 2

STAT 801A, Fall 2026

Due on January 1, 2026 at 9:00 pm

Part 1: `z.test()` function

Although many statistical courses use a z-test, R does not have a built in `z.test()` function. Your task is to create and use a `z.test()` function for all three cases of hypothesis tests.

1. Create a `z.test` function using the template provided below.

```
z.test <- function(y, mu = 0, sigma = 1,
                  alternative = c('two.sided', 'less', 'greater'),
                  conf.level = 0.95) {

  # Calculations here

  # Organize results
  results <- list(
    data = y,
    z_c = ,
    z_r = ,
    z_c_in_rejection_region = ,
    p.value = ,
    alternative =
  )

  # Output list of results
  return(results)
}
```

2. Use the data in `z-test-data.csv` and your `z.test()` function to test the following hypothesis. Use $\alpha = 0.05$. For each hypothesis test, include the output of your function

and a written interpretation of the conclusion (i.e., what can you conclude about the data provided the results of the test?).

- a. Column prob2a – $H_0 : \mu = 30$ vs. $H_A : \mu > 30$
- b. Column prob2b – $H_0 : \mu = 15$ vs. $H_A : \mu < 15$
- c. Column prob2c – $H_0 : \mu = 50$ vs. $H_A : \mu \neq 50$

Part 2: one sample t-test

Each summer, Nitrogen from the Mississippi River Basin is believed to be partly responsible for the large zone of oxygen-depleted water that develops in the Gulf of Mexico. One such river basin that drains into the Mississippi River is the Minnesota River. Between 1980 and 1996, the mean Nitrogen (in mg/L) in the Minnesota River was estimated to be 4.19. Suppose that a recent data collection effort obtained 30 samples, provided in `mn-river-n-samples.csv`. However, two samples were accidentally destroyed in transit. Researchers are interested in if Nitrogen increased in the Minnesota River.

Data was inspired by the following article: [https://doi-org.libproxy.unl.edu/10.1016/S0048-9697\(99\)00532-X](https://doi-org.libproxy.unl.edu/10.1016/S0048-9697(99)00532-X)

```
set.seed(218)
N_mg_l <- round(rnorm(n=30, mean = 4.3, sd = 1/6),2)
ID <- 1:30
N_mg_l[c(13, 25)] <- NA
mn <- data.frame(ID, N_mg_l)
write.csv(mn, file = 'mn-river-n-samples.csv', row.names = F)
```

3. Using `qqnorm()` and `qqline()`, and one other graphical method of your choice, describe whether the
4. Define the appropriate null and alternative hypotheses that satisfies the research question.
5. Conduct a one-sample t-test without using the `t.test()`.
6. Conduct a one-sample t-test with the `t.test()` function.
7. Write a conclusion (1-2 sentences) that satisfies the research question.